MIL-R-43031E
2 November 1977
SUPERSEDING
MIL-R-43031D
31 March 1970

MILITARY SPECIFICATION

REFRIGERATION UNITS, MECHANICAL, PANEL TYPE,

3000 BTU/HR (879 W) CAPACITY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers panel type, mechanical refrigeration units having a capacity of 3000 British thermal units per hour (Btu/hr) (879 W).

1.2 Classification. The refrigeration units covered by this specification shall be of the following types as specified (see 6.2).

Type I - Gasoline engine driven compressor

Type II - Electric motor driven compressor

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command, Natick, MA 01760, by using the self-addressed standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4110

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SPECIFICATIONS

FEDERAL

O-F-506	-	Flux, Soldering; Paste and Liquid
P-D-680	-	Dry Cleaning Solvent
BB-F-1421	-	Fluorocarbon Refrigerants
CC-M-1807	-	Motor, Alternating Current, Fractional and Integral Horsepower, (500 hp and Smaller)
QQ-A-250/2	-	Aluminum Alloy, 3003, Plate and Sheet
QQ-A-250/8	-	Aluminum Alloy, 5052, Plate and Sheet
QQ-S-654	-	Brazing Alloy, Silver
QQ-S-571	-	Solder; Tin Alloy; Tin-Lead Alloy; and Lead Alloy
QQ-S-698	-	Steel, Sheet and Strip, Low Carbon
QQ-S-700	-	Steel, Sheet and Strip, Medium Carbon and High Carbon
TT-C-490	-	Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings
TT-E-529	-	Enamel, Alkyd, Semi-Gloss
TT-P-1757	-	Primer Coating, Zinc Chromate, Low Moisture Sensitivity
VV-L-825	-	Lubricating Oil, Refrigerant Compressor
WW-T-775	-	Tube, Copper, Seamless, (For Refrigeration and General Use)

MILITARY

MIL-V-173	-	Varnish, Moisture and Fungus-Resistant (for the Treatment of Communications, Electronic, and Associated Electrical Equipment)
MIL-P-514	-	Plates, Identification, Instruction and Marking, Blank
MIL-F-3541	-	Fittings, Lubrication
MIL-R-3593	-	Refrigeration and Cooling Equipment (Excluding Household Refrigerators) Packaging Of
MIL-W-5086/1	-	Wire, Electrical, Hookup and Interconnection, Polyvinyl Chloride Insulation, Nylon Jacket, Tin-Coated Copper Conductor, 600 Volt, 105°C
MIL-C-5541	-	Chemical Films for Aluminum and Aluminum Alloys
MIL-C-10578	-	Corrosion Removing and Metal Conditioning Compound (Phosphoric Acid Base)
MIL-S-12514	-	Starter, Motor, Across-the-Line and Reduced Voltage Types A-C Induction, 3/4 to 100 hp Manual and Electric
MIL-E-14328	-	Rubber, Synthetic, Medium Soft, General Purpose Gasket Material (For Extreme Climatic Conditions)
MIL-R-43024	-	Refrigerator Mechanical, Commercial, Portable, Reach-In 70 Cubic Foot - Less Mechanical Unit
MIL-T-46786	-	Tank, Fuel, Engine: General Requirements For

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STANDARDS

FEDERAL

FED-STD-595 - Colors

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130 - Identification Marking of US Military Property
MIL-STD-279 - Compressor, Reciprocating Power Driven, Open Type, for Use with Refrigerant-12
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of
MIL-STD-759 - Condensers, Air Cooled, Refrigerant-12
MIL-STD-889 - Dissimilar Metals
MIL-STD-1227 - Engine, Gasoline: Air Cooled, 3 BHP, 4-cycle, Military Design, Model 2A016, Installation Procedures
MIL-STD-1474 - Noise Limits for Army Material
MS 16993 - Plug, Fusible, Refrigerant-12
MS 17241 - Heat Interchanger, Refrigerant-12
MS 17243 - Valves, Angle, Shut Off, Packed, Receiver, Refrigerant-12
MS 17291 - Receivers, Liquid, Refrigerant-12
MS 17415 - Valve, Safety Relief, Refrigerant-12
MS 17798 - Indicator, Sight, Liquid, Refrigerant
MS 17982 - Valves, Expansion, Thermostatic, Refrigerant-12 and Refrigerant-22
MS 27211 - Thermometer, Indicating, Capillary Tube and Bulb, 2-1/2 Inch Dial Size
MS 27900 - Gages, Pressure and Gages Compound, Pressure and Vacuum, Dial Indicating Panel Mounted, 2-1/2 Inch Dial Size, Refrigerant R-12 and R-22
MS 33586 - Metals, Definition of Dissimilar
MS 35845 - Dehydrators, Desiccant, Refrigerant-12
MS 35867 - Connection End, Internal Flare, Refrigeration
MS 35868 - Connection End, External Pipe Thread Refrigeration
MS 35869 - Adapter, Straight, Male, Refrigeration Pipe to Flared Tube
MS 35871 - Tees, Pipe to Tube, Refrigeration
MS 35872 - Flare Nut and Flare Seal Bonnet, Refrigeration
MS 35880 - Valves, Packless Stop, Solder-Joint Connection Ends
MS 35881 - Valves, Packless, Stop, Flare Connection Ends

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MS 35917	-	Elbows, Tube, 90 and 45 Degree, Solder-Joint, Refrigeration
MS 35918	-	Adapter, Straight, Pipe Thread to Tube, Male Solder-Joint Refrigeration
MS 35919	-	Adapter, Straight, Pipe to Tube, External Flare to Female Solder-Joint, Refrigeration
MS 35920	-	Reducer, Coupling, Tube, Male to Female, Solder-Joint, Refrigeration
MS 35921	-	Adapter, Straight, Pipe to Tube, Pipe Thread to Female Solder-Joint, Refrigeration
MS 35922	-	Elbow, Tube, 90 Degree, Reducing, Solder-Joint, Refrigeration
MS 35923	-	Reducer, Coupling, Tube, Female to Female, Solder-Joint, Refrigeration
MS 35924	-	Plug and Cap, Tube, Solder-Joint, Refrigeration
MS 35925	-	Connection End, Solder-Joint, Refrigeration
MS 35926	-	Connection End, Internal Pipe Thread, Refrigeration
MS 35927	-	Coupling, Tube, With Stop, Solder-Joint, Refrigeration
MS 35928	-	Elbows, Tube, Street, 90 and 45 Degree, Solder-Joint Refrigeration
MS 35929	-	Tees, Tube, Solder-Joint, Refrigeration

DRAWING

US ARMY NATICK RESEARCH AND DEVELOPMENT COMMAND

5-13-1973	-	Refrigeration Unit, Panel Type, 3000 BTU/HR Capacity, for Refrigerator Portable, Reach-In 70 Cu. Ft.

NOTE: Miniature copy of Drawing 5-13-1973, identified as figure 1 is attached for information purposes only.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

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NATIONAL BUREAU OF STANDARDS

Handbook H-28 - Screw Thread Standards for Federal Services
Product Standard PS-1 - Softwood Plywood, Construction and Industrial

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

Specification S 1.4 for General Purpose Sound Level Meters

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

UNDERWRITERS LABORATORIES, INC. (UL)

UL Standard 207 (c) - Refrigerant-Containing Components

(Application for copies should be addressed to the Underwriters Laboratories, Inc., 207 East Ohio Street, Chicago, IL 60611; 333 Pfingsten Road, Northbrook, IL 60062; 1285 Walt Whitman Road, Melville, L.I., New York, NY 11746; 1655 Scott Boulevard, Santa Clara, CA 95050 or 2602 Tampa East Blvd., Tampa, FL 33619.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

* 3.1 First article. When specified (see 6.2), the contractor shall furnish a sample of the item to be procured under this specification for first article inspection and approval (see 4.3 and 6.3).

* 3.1.1 Initial production testing sample. When specified (see 6.2), the contractor shall furnish a sample for initial production inspection and approval (see 4.3.1 and 6.9).

* 3.1.2 Inspection comparison testing sample. When specified (see 6.2), the contractor shall furnish a sample for comparison inspection and approval (see 4.3.2 and 6.10).

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3.2 Standard product. Each component, including accessories, instruments, and controls, except Military standard items, shall be a manufacturer's current standard catalog product except for any design modifications and additional requirements specified herein.

* 3.3 Codes and standards. The refrigeration components shall comply with the applicable requirements of UL Standard 207 (c).

* 3.3.1 Compliance. Prior to approval of the first article, if one is submitted, or prior to commencing production, the contractor shall submit to the contracting officer or his authorized representative satisfactory evidence that the refrigeration unit components, as applicable, he proposes to furnish under this specification, meet the applicable requirements of UL Standard 207 (c) as follows:

* 3.3.1.1 UL. Acceptable evidence of meeting the requirements of UL shall be the UL label, a UL listing mark, or a certified test report from a recognized independent testing laboratory, acceptable to the Government, indicating that the refrigeration components have been tested and conform to UL Standard 207 (c).

* 3.4 Materials and components. Materials not definitely specified shall be of the quality normally used by the manufacturer for refrigeration units provided the completed item complies with all provisions of this specification. Components, including accessories, instruments, and controls, except Military standard items, shall be a current standard catalog product of its manufacturer except for any modifications required to meet specification requirements. Commercial tolerances shall apply to all metal thicknesses and diameters unless otherwise specified (see 6.11).

3.4.1 Lumber. All lumber shall be select grade hardwood.

3.4.2 Plywood. All plywood shall be group I, exterior type, grade B-C shall conform to PS 1.

3.4.3 Preservative, wood. Wood preservative shall conform to P-D-680 and shall contain one percent solubilized copper-8-quinolinolate free of amines and napthenic acid or its derivatives.

3.4.4 Fastening devices. Steel bolts, nuts, screws, and washers for the evaporator shall be zinc-plated, and on sections other than the evaporator shall be zinc or cadmium plated. Brass bolts, nuts, washers, and machine screws shall not be used in fabricating aluminum parts. Threads shall conform to National Bureau of Standards Handbook H28.

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3.4.5 Gasket. The gasket for the heat shield shall be fabricated of resilient sponge rubber core incased in an oil resistant synthetic rubber skin and of the cross-section dimensions shown on Drawing 5-13-1973. The gasket core may be either natural or synthetic rubber. The minimum thickness of the skin shall be 0.015 inch (0.38 mm).

3.4.6 Metal conditioner.

3.4.6.1 Ferrous metal conditioner. Surface conditioner, applied before painting ferrous metals, shall conform to type I or II, method VI of TT-C-490 or type I or type II compound of MIL-C-10578.

3.4.6.2 Aluminum metal conditioner. Aluminum surface conditioner shall conform to MIL-C-5541.

3.4.7 Metal primer. Primer shall conform to TT-P-1757.

3.4.8 Finish paint. The finish paint shall be semi-gloss enamel conforming to TT-E-529. Unless otherwise specified (see 6.2), the color shall conform to No. 24087 of FED-STD-595.

3.4.9 Refrigerant. The refrigerant shall conform to type 12 of BB-F-1421.

3.4.10 Aluminum alloy sheet or strip. Aluminum alloy sheet or strip shall be alloy 5052, temper H32 conforming to QQ-A-250/8, or alloy 3003 temper H14 conforming to QQ-A-250/2.

3.4.11 Steel sheet and strip. Steel sheet or strip shall conform to QQ-S-698 and QQ-S-700.

3.4.12 Silver solder. Silver solder shall conform to QQ-B-654.

3.4.13 Solder. Solder shall conform to composition Sb5 of QQ-S-571.

3.4.14 Soldering flux. Soldering flux shall conform to type I of O-F-506.

3.4.15 Insulation. Insulation shall have a manufacturer's commercial rated K factor of not over 0.15 British thermal units per hour (Btuh) per square foot per inch of thickness per degree Fahrenheit (F) temperature difference at a mean temperature of 75°F (24°C) (0.02 W/m.K). Insulation shall be odor free, fire retardant and self-extinguishing.

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3.4.16 Compressor lubricating oil. Compressor lubricating oil shall conform to type II of VV-L-825 or may be oil recommended by the compressor manufacturer.

3.4.17 Pressure lubricating fittings. Pressure lubricating fittings shall conform to type I of MIL-F-3541.

3.4.18 Tube, copper. Tube used for refrigerant lines shall conform to WW-T-775.

3.5 Design and construction. All components and assemblies of the refrigeration units shall conform to the requirements specified herein and the overall and mounting dimensions shall conform to Drawing 5-13-1973. The weight of the refrigeration units shall not exceed 500 pounds (226 kg). The principal components shall be a condensing unit with a gasoline-engine-driven compressor for the type I units, an electric-motor driven compressor for the type II units, air cooled condenser, forced-air evaporator section, controls, and a framed inclosure. All mechanical equipment and controls shall function as intended without failure when tested as specified in 4.6.1. The refrigeration system shall be designed to use the refrigerant specified in 3.4.9. An insulated panel shall be provided to mount the unit to a refrigerator and to provide a heat shield between the evaporator and condenser sections. Defrosting of type I units shall be accomplished by a manually controlled hot gas system. Defrosting of type II units shall be accomplished by an automatic hot gas system. The refrigeration units shall not be damaged or distorted and shall remain operable after being raised 1 inch (25 mm) and dropped onto a concrete surface fifty times while operating (see 4.6.6). The portion of the refrigeration unit exposed to weather shall be capable of operating in rainfall up to 3 inches (76 mm) per hour without functional damage or impairment of operation (see 4.6.7). There shall be free access to inspect or service and adjust any components or equipment, as required, for proper operation of the unit. All parts of the refrigeration unit, with the exception of the compressor, engine or motor, and clutch shall be interchangeable with part of other units furnished on any one contract. The compressor, engine or motor, and clutch shall be interchangeable as complete assemblies. The compressor, motor, or engine shall be mounted on a flat steel base fabricated from not less than 1/4-inch (6 mm) thick steel. The steel base shall be attached to the unit frame by appropriately sized shock mounts designed to prevent transmission of vibratory shock to the unit. All piping and other connections made to the compressor, condenser, or receiver shall be designed to withstand flexing to which they will be subjected to during normal operation and referenced tests. All electrical wiring shall be in accordance with type II of MIL-S-5086 and shall be neatly arranged and supported, to prevent sagging and chafing. All references of the ambient air temperatures shall be considered to mean the average temperature of the air entering the condenser. The completed refrigeration unit shall be capable of withstanding storage temperature of 155°F (68°C) for 4 hours and -65°F (-54°C) for 12 hours without causing permanent set or loss of resiliency of packing and gaskets, discoloring, cracking, bulging, checking, or crazing when tested as specified in 4.6.13.

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3.5.1 Conversion kit. The type I refrigeration unit shall be designed to permit conversion from gasoline engine compressor drive to an electric-motor compressor drive by means of a conversion kit (see 4.6.9 and 6.7). The contractor shall furnish the modification procedure, wiring diagram, and a list of all conversion kit component parts. The conversion kit shall include but not be limited to a 3 horsepower (hp) (2 kw) electric motor (see 3.5.4.6). The conversion kit shall include the applicable sheave to maintain the same compressor speed as when operating on gasoline-engine-drive, and electric starter (see 3.5.4.6.1), an off-on switch mounted in the starter box cover, and a high pressure cutout switch with automatic reset and set for 250 psi (1723 kPa).

3.5.2 Performance characteristics.

3.5.2.1 Capacity. The refrigeration units shall meet the capacity requirements specified herein (see 4.6.3).

3.5.2.1.1 Cooling capacity at 0°F (-18°C) refrigerator temperature. The low temperature net capacity for the refrigeration units exclusive of evaporator fan heat gain, shall be not less than 3,000 Btu per hour (879 W) with the compressor discharge pressure at no time exceeding 200 pounds per square inch (psi) (1378 kPa). The capacity shall be obtained when the air entering the condenser is at a temperature of 110°F (43°C) ± 2°F (± 1°C); the temperature of the air entering the evaporator is 0°F (-18°C) ± 1°F (± 0.5°C); the temperature of the air leaving the evaporator is minus 4°F (-20°C), ± 1°F (± 0.5°C); and the refrigerant temperature, considered as refrigerant saturation temperature corresponding to the pressure at the evaporator outlet, is not lower than -12°F (-24°C).

3.5.2.1.2 Cooling capacity at 35°F (2°C) refrigerator temperature. The high temperature net capacity for the refrigeration units exclusive of the evaporator fan heat gain, shall be not less than 4500 Btu/hr (1318 W) with the compressor discharge pressure at no time exceeding 200 psi (1378 kPa). The capacity shall be obtained when the air entering the condenser is at a temperature of 110°F (43°C) ± 2°F (± 1°C); the temperature of the air entering the evaporator is 35°F (2°C) ± 1°F (± 0.5°C) and the refrigerant temperature considered as refrigerant saturation temperature corresponding to the pressure at the evaporator outlet is not lower than 12°F (-11°C).

3.5.2.2 Noise level. When tested as specified in 4.6.14, the noise limits for all types of refrigeration units shall conform to MIL-STD-1474.

3.5.3 Refrigeration unit inclosure.

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3.5.3.1 Unit inclosure. The refrigeration unit inclosure shall be as shown on Drawing 5-13-1973. The inclosure frame shall be fabricated of 0.074 inch (2 mm) thick steel sheet or angle specified in 3.4.11 or 0.100 inch (2.5 mm) thick aluminum alloy sheet specified in 3.4.10. Frame members shall be joined by welding. The front and sides of the inclosure shall be inclosed with hinged doors and the top shall be covered with a removable panel. The doors shall have a minimum 1-1/2 inch (38 mm) border. The panels shall have a minimum 1-inch (25 mm) border. The borders shall be of the same metal and thickness as the panel or door. The panel or door shall be flattened, expanded aluminum or aluminum alloy, or flattened, expanded steel. The expanded metal shall have a maximum hole dimension of 1-13/16 inch (46 mm). The refrigeration unit housing, including panels, door frames, and expanded metals, shall be aluminum or aluminum alloy having a thickness of 0.063 inch (1.6 mm) or steel having a thickness of 0.059 inch (1.5 mm). Hinged doors shall be secured with barrel bolts not less than 2-1/2 inches (63.5 mm) nor more than 4 inches (102 mm) in length. Two steel "D" rings, shall be permanently attached to the top of the framework as shown on Drawing 5-13-1973. The "D" rings shall be located in the same vertical plane as the center of gravity of the unit. Each "D" ring shall be capable of supporting the entire weight of the refrigeration unit and shall be tested for location and strength as specified in 4.6.8. Aluminum or aluminum alloy plate 0.100 inch (2.5 mm) thick shall cover the bottom of the unit inclosure.

3.5.3.2 Heat shield. An insulated heat shield shall be provided to separate and insulate the evaporator from the condensing unit. The heat shield shall be constructed from lumber specified in 3.4.1, or plywood specified in 3.4.2 or both. After wood and plywood members have been cut to a size and completely milled, they shall be immersed in wood preservative specified in 3.4.3 for not less than 30 minutes and air dried for a minimum of 24 hours. The condenser side shall be metal covered using the same material and thickness as the refrigeration unit inclosure (see 3.5.3.1) and the evaporator side shall be covered with aluminum or aluminum alloy having a thickness of 0.050 inches (13 mm). All joints shall be secured with bolts or screws. All space between sheathing, if used, framing, and metal covering shall be completely filled with 3-1/2 inches (89 mm) of insulation specified in 3.4.15.

The insulation shall be installed so as to hold itself in place and give a good vapor seal. No metal or other high heat conductor shall extend through the insulation except for the fan shaft, refrigerant piping, condensate drain, and electrical wiring. All holes extending through the heat shield shall be sleeved or grommeted to prevent abrasion between the piping or wiring and the metal skin. The holes shall be sealed with rubber or asphaltic-base material compound for use without heating. An arrow indicating the direction of the rotation of the electric motor shall be placed on the heat shield near the compressor or prime mover.

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3.5.3.3 Guards. Guards for protection of the evaporator and condenser coils shall be provided and shall be fabricated of flattened expanded metal retained within a 1-inch (25 mm) minimum metal border. The guards shall be removable without removing or loosening any other component except fastening devices. The evaporator guard shall be aluminum or aluminum alloy having a thickness of 0.063 inch (1.6 mm). The condenser guard shall be of steel having a thickness of 0.059 inch (1.5 mm).

3.5.3.4 Control panel. A control panel shall be mounted and contained within the unit framework at a convenient location. The panel on type I units shall contain the defrost control valve, the damper control, the compound and pressure gages, the prime mover controls, the dial thermometer, and the high temperature control valve. The panel on type II units shall contain the on-off switch, the compound and pressure gages, the dial thermometer, the thermostat and the defrost indicator light.

3.5.4 Refrigeration system.

3.5.4.1 Refrigerant piping. All tubes shall conform to the material specified in 3.4.18. Minimum sizes of copper tube with related valve and fittings shall be as follows: 5/8-inch (16 mm) O.D., suction line, 3/8-inch (9.5 mm) O.D., liquid line, and a 1/2-inch O.D., discharge line, condenser to receiver line and hot gas line. Valves and fittings shall conform to MS 16993, MS 35867 through MS 35869, MS 35871, MS 35872, MS 35880, MS 35881, and MS 35917 thru MS 35929, as applicable. Connections shall be solder-joint type, except that flared connections shall be used on the expansion valve, hot gas by-pass valve, dehydrator, gages, and receiver valves. Piping shall be arranged to resist vibration. Particular attention shall be given to compressor connections. Vibration eliminators shall be installed in piping connections to the compressor. The flexible length of the vibration eliminators shall not be less than 7 inches (178 mm). All solder connections shall be made with solder specified in 3.4.13, except that brazing may be used whenever brazing temperatures will not be deleterious to any fitting or component adjacent to the brazed joint. All flux and foreign material shall be removed from finished joints, and tube and fittings shall be given a coat of protective lacquer after successfully passing the leak test. All refrigerant containing components and piping shall be gas tight (see 4.5.11). All components to be soldered or brazed, having internal parts that could be damaged during this process, shall be disassembled before soldering or brazing.

3.5.4.2 Compressor. The compressors shall conform to type I of MIL-STD-279. Compressors shall be installed to permit observation of the crankcase oil level. Compressor mounting bolts shall be 3/8 inch-16 UNC and the base shall have captive nuts.

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3.5.4.3 Gages, compound and pressure. The compound and pressure gages shall conform to MS 27900, except that 1/4-inch (6 mm) flare connections shall be acceptable. Gages shall be properly identified and shall be mounted on the front of the control panel and connected to the back seating connections of the compressor shut-off valves.

3.5.4.4 Power transmission. Power shall be transmitted from the gasoline engine for type I units, and the electric motor for the type II units, to the compressor through a minimum of two A-section V-belts with belt adjustment made by movement of the prime mover by means of tension or compression bolts. Type I units, the evaporator and condenser fans shall be driven from the compressor employing an A-section V-belt with belt tension adjustment provided by an adjustable idler sheave. The idler sheave bearings shall be of the pressure lubricated type. The evaporator and condenser fans on the type II units shall be driven from a separate fan motor an an "A" section V-belt. Belt tension shall be adjusted by movement of the fan motor. All belts shall be a standard catalog length of at least three belt manufacturers. All sheaves shall be designed for operation with either A- or B- section belts. All belts shall be replaceable without removal of any major component of the refrigeration unit. The gasoline engine drives shall have a centrifugal-clutch sheave.

3.5.4.5 Gasoline engine for type I refrigeration units. The gasoline engine model 2A016 shall be Government furnished property (see 6.2 and 6.8). The engine shall be installed as specified in MIL-STD-1227. The engine oil temperature shall not exceed 250°F (121°C) and the cylinder head temperature shall not exceed 475°F (246°C) when tested as specified in 4.6.5.1.

3.5.4.5.1 Fuel tank. A 6-gallon (22 L) capacity fuel tank in accordance with type I of MIL-T-46786 shall be furnished and installed. The mounting shall be such that neither vibration, excessive tightening of mounting, nor contact with other members of the engine or end item will cause leaks to develop. A chain-secured type filler cap, a filler neck, and a strainer with a clear opening of not less than 2 inches (51 mm) in diameter, all in conformance with item 1, table II of MIL-T-46786 shall be provided. The removable strainer shall be located or protected in such a manner as to prevent contact with the filling nozzle. The filler neck shall be provided with an air vent to prevent "blow back" on rapid filling and a drain fitting to enable draining without removing the tank from the unit. The fuel tank shall be located at the bottom of the refrigeration unit, the bottom of the fuel tank shall be not lower than 32 inches (813 mm) below the bottom of the carburator. The intake of the fuel supply system of the engine shall not be more than 1/4 inch (6.3 mm) above the bottom of the fuel tank. The fuel tank shall be protected by a 0.100 inch (2.5 mm) thick aluminum or aluminum alloy sheet material specified in 3.4.10 covering the bottom of the unit. The aluminum or aluminum alloy sheet covering the bottom of the unit shall have a hole to permit draining of the fuel tank. The fuel supply system shall be as shown on Drawing 5-13-1973 and shall not exceed 36-inches (914 mm) in length, including the primer.

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3.5.4.5.1.1 Gasoline gage. A mechanical gasoline gage for type I units, to indicate the gasoline level in the fuel tank, shall be mounted on the fuel tank so as to be easily visible.

3.5.4.5.2 Muffler and exhaust pipe. An exhaust muffler shall be furnished and installed for the type I refrigeration units. The engine exhaust manifold shall be connected to the muffler by flexible corrosion-resisting steel tube having a length not less than eight times the diameter of the tube. The muffler shall be rigidly mounted to the upper part of the frame of the unit. The back pressure in the exhaust system shall not exceed 8 inches (203 mm) water column when measured at the engine manifold while the engine is running at the full load required by the unit (see 4.6.5.1). The muffler shall be chrome impregnated or diffused corrosion-resisting steel or aluminized steel having a thickness of 0.06 inch (1.5 mm). The exhaust shall be protected against the entry of rain and shall include drainage provisions to prevent accumulated water and condensed vapors.

3.5.4.6 Electric motor. Unless otherwise specified (see 6.2), a 3 horsepower (0.2 w) nominal 1750 revolutions per minute (rpm) electric motor shall be provided to operate from a nominal 208 volt, 3 phase, 60 hertz power supply. The motor shall be of the squirrel-cage type NEMA Design "B" with a NEMA 182T frame, designed for continuous duty, protected by a drip proof enclosure, and shall conform to CC-M-1807.

3.5.4.6.1 Motor starter. A general purpose motor starter conforming to type I, class 2, size 1, of MIL-S-12514 shall be mounted in a drip tight enclosure except that a toggle switch shall be provided instead of a push-button station. The switch shall provide for the selection of "ON" and "OFF". Fuses are not acceptable in lieu of the thermal overload protective device.

3.5.4.7 High pressure protection. For the type I units an automatic re-seating type safety relief valve, conforming to MS 17415 set at 300 pounds per square inch (psi) (2068 kPa), shall be installed between the compressor discharge line and hot gas line to the evaporator at a point downstream of all valves in the hot gas line. The type II units shall be supplied with a high pressure cutout switch, having an automatic reset and set for 250 psi (1723 kPa).

3.5.4.8 Condenser. The condenser shall be an air-cooled type conforming to class I, type V of MIL-STD-759. The condenser fan shall conform to MIL-STD-759. The fan shaft shall be as specified in 3.5.4.11.2. The condenser shall be removable from the front of the unit.

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3.5.4.9 Receiver. The receiver shall conform to part No. MS 17291-2 of MS 17291, and shall be fitted with a fusible plug conforming to part No. MS 16993-2 of MS 16993, and inlet and outlet valves shall conform to MS 17243, except they shall be sized to fit the lines.

3.5.4.10 Thermometer. A dial type thermometer conforming to part No. MS 27211-2 of MS 27211 shall be mounted on the control panel with the remote bulb secured to the air intake side of the evaporator to sense the temperature of the entering air.

3.5.4.11 Evaporator. The evaporator section shall be mounted on the heat shield and shall consist of the evaporator coil, fan, thermostatic expansion valve, and heat interchanger. The evaporator shall be of sufficient capacity to meet the refrigeration capacities specified in 3.5.2.1.1 and 3.5.2.1.2, and shall utilize hot gas defrosting. The evaporator shall be a multi-pass, forced convection type made of copper tube not less than 0.016 inch (0.4 mm) wall thickness, for 3/8-inch (9.5 mm) and 1/2-inch (13 mm) O.D. tubing, and 0.020 inch (0.05 mm) wall thickness for 5/8-inch (16 mm) O.D. tube, expanded into aluminum fins. Fins shall be 0.009 inch (0.2 mm) thick and shall be spaced not more than six per inch (25 mm). Tube sheets shall be constructed of aluminum or aluminum alloy. The evaporator coil shall be equipped with a multiple outlet type distributor and a suitable defrost header or distributor. The housing shall be constructed of aluminum or aluminum alloy, thickness of 0.100 inch (2.5 mm). An aluminum or aluminum alloy drip pan with a hot gas defrost drain pan heater shall be provided to collect condensate. Condensate shall flow through a drain in the heat shield to a trap located outside the heat shield. No paint shall be used on the evaporator section.

3.5.4.11.1 Evaporator fan. The evaporator fan shall be a dynamically balanced centrifugal blower or statically balanced fan mounted on a shaft extending through the insulated heat shield. The fan shall be constructed of steel treated to resist corrosion, corrosion resistant metal, or a combination of these materials. The fan shall be designed to draw air through the evaporator with air discharge at the top third of the evaporator housing.

3.5.4.11.2 Fan drive. A common shaft shall be used for the condenser and evaporator fans and shall be 1-inch (25 mm) diameter and shall be step machined with a minimum 1/16-inch radius to accept the fans. Two, 1-inch (25 mm), pressure lubricated, 4-bolt flange type, ball bearings shall be used to support the fan shaft.

3.5.4.11.3 Fan motor for type II units. The fan motor for type II units shall be a 1/4 hp, 1750 rpm motor for operation on 208 volt, 60 Hz, single phase power. The motor shall be totally enclosed, fan cooled (TEFC) and shall have a NEMA 48 frame. The motor shall be equipped with automatic reset overload protection. The fan motor shall be wired to operate when the compressor motor operates, except during defrosting.

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3.5.4.12 Thermostatic expansion valve. A thermostatic expansion valve selected from MS 17982 shall be provided.

3.5.4.13 Heat interchanger. A heat interchanger conforming to part No. 4 of MS 17241 shall be installed in the evaporator section.

3.5.4.14 Defrost system. The defrost system shall be capable of operation through a complete defrost cycle and return to a normal refrigeration cycle without overloading the prime mover, without splashing or dripping water inside the refrigerator, without damaging the compressor because of lack of lubrication or because of refrigerant slugging when tested as specified in 4.6.4. The defrost system shall reclaim not less than 3-3/4 pounds (1.7 kg) of condensate within 30 minutes when tested as specified in 4.6.4. A non-corrodible instruction plate with complete instructions for defrosting the refrigeration unit shall be attached near the control panel (see 3.7.2).

3.5.4.14.1 Defrost valve for type I units. The defrost valve for the type I units shall be a manually operated line valve installed in a by-pass around the control line valve and thermomechanical valve specified in 3.5.4.15.

3.5.4.14.2 Damper for type I units. Manually operated damper for type I units shall be located on the downstream side of the evaporator. An evaporator air damper control handle shall be located on the control panel or in a location close to the defrost valve. The damper shall be closed during operation of the defrost system to keep warm air from circulating in the refrigerator.

3.5.4.14.3 Defrost control type II. The defrost of the type II units shall be automatic and shall be initiated by a time clock and terminated by a thermostat located at the outlet of the evaporator coil. The time clock shall be adjustable for one to three defrost cycles per day. The clock shall open a defrost solenoid and shut-off the fan motor during defrost. An indicator light shall be furnished on the control panel to indicate when the unit is defrosting.

3.5.4.15 Temperature control. Refrigeration temperature control for type I units shall be accomplished by load by-passing as required. The control system shall consist of a hot gas line to the evaporator controlled by a manual line valve and a thermomechanical valve with its thermal element located in the inlet air stream of the evaporator. With the manual line valve closed, the system shall provide for subfreezing refrigerator temperatures and when in the open position shall permit the thermomechanical valve to control the box temperature at 35°F (1.6°C) or above as desired by by-passing hot gas automatically. The type II units shall control temperature by stopping and starting the unit using a thermostat. The thermostat shall have a range of minus 10°F (-12°C) to plus 50°F (10°C). The thermostat shall be mounted on the control panel.

3.5.4.16 Dehydrator. A dehydrator conforming to MS 35845 shall be installed in the liquid line between the receiver and the expansion valve. A three valve by-pass shall be installed in the refrigerant line to facilitate replacement of the dehydrator and the receiver outlet valve shall be one of the three valves.

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3.5.4.17 Liquid sight and moisture indicator. A liquid sight and moisture indicator selected from MS 17798 shall be installed in the liquid line.

3.5.4.18 Charging refrigeration system. Each complete refrigeration unit shall be evacuated and dehydrated, and fully charged with the refrigerant specified in 3.4.9 and the oil specified in 3.4.16. The moisture content of the refrigerant shall not exceed 25 parts per million (ppm) when tested as specified in 4.5.10.

3.5.4.19 Fungus control of coils and electric motors. Starter coils and solenoid coils, except potted coils, shall be treated in accordance with MIL-T-152. Motor coils with a temperature rise of 40°C or less shall be coated with fungus resistant varnish conforming to MIL-V-173. The coats shall be applied by the vacuum-pressure, immersion, centrifugal, pulsating pressure, or build-up method to fill all interstices in the coils and preclude entrapped air or moisture. The sealer coat may also be applied by brushing or spraying.

3.5.4.20 Finish.

3.5.4.20.1 Surface preparation. The exterior and interior surfaces of the framework inclosure, base and power unit except where not desirable for proper operation, shall be cleaned and conditioned. Rough or sharp edges and corners, burrs, and slivers shall be removed. Ferrous surfaces on any components to be painted shall first be thoroughly cleaned and conditioned using the compound specified in 3.4.6.1. Wood surfaces shall be unfinished after being preservative-coated. Aluminum and aluminum alloy surfaces shall be cleaned and treated by conditioner specified in 3.4.6.2.

3.5.4.20.2 Priming paint. After surfaces are thoroughly dried from cleaning and conditioning, a coat of primer shall be applied conforming to 3.4.7.

3.5.4.20.3 Finish coat. After primer has thoroughly dried, surfaces shall be given a coat of enamel specified in 3.4.8. Enamel shall be applied so that there will be no streaks, crops, pinholes, checks, or cracks. Care should be taken not to paint or allow paint to get on the fins of the condenser or the heat shield gasket or any other portion of the unit where paint would be detrimental, or on any portion of the unit extending into the insulated structure.

3.5.4.21 Lubrication.

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3.5.4.21.1 Lubricating. Lubricants for meeting test, operating, and storage requirements of this specification shall be used to lubricate wearing parts of this equipment. Particular care shall be exercised in lubricating the gasoline engine and accessories and the evaporator and condenser blowers or fans. All points requiring lubrication shall be lubricated before testing. Units run over 50 hours shall be relubricated before shipping.

3.5.4.21.2 Lubricating fittings. Unless otherwise specified (see 6.2) all bearing surfaces shall be equipped with pressure lubrication fittings specified in 3.4.17, and the fittings shall be located to be easily accessible from the front of the units. Each fitting shall be enameled bright red.

3.5.4.21.3 Tagging lubrication fittings. After lubrication, each item of equipment shall have the fittings or location tagged to indicate the temperature range and grade of lubricant used and recommended frequency of lubrication.

3.5.4.21.4 Lubrication precaution marking. Where the use of high pressure lubrication equipment, for example, 1,000 psi (6894 kPa) or higher, will damage grease seals or other parts, a suitable warning plate to this effect shall be located at the appropriate place or places. Any other lubrication precautions should likewise be included on a warning plate.

3.5.4.21.5 Lubrication instructions. Lubrication instructions for performing periodic preventative maintenance as specified (see 6.2), shall be affixed to each refrigeration unit in a convenient location.

3.6 Electromagnetic compatibility. Equipment procured shall be designed and equipped for electromagnetic compatibility in accordance with class II B of MIL-STD-461 (see 4.6.12).

3.7 Identification.

3.7.1 Identification plate. The identification plate shall conform to type III, composition A, class 1 or 2 of MIL-P-514. The refrigeration units shall be marked in accordance with MIL-STD-130 and with the following:

(a) Nomenclature
(b) Type of unit, capactiy of unit, and type of refrigerant and charge
(c) National stock number
(d) Contract number
(e) Military model number
(f) Manufacturer's name, model, and serial number
(g) Manufacturer's address
(h) US

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3.7.2 Instruction plates. Instruction plates made of nonferrous metal shall be attached to each refrigeration unit. The instruction plates shall have the wiring diagram and refrigeration cycle diagram. The plates shall contain instructions for startup, operating, defrosting, shutdown, and preventive maintenance procedures. Lettering shall not be less than 1/16 inch (1.5 mm) high.

3.7.3 Control and gage nameplates. All control switches and gages shall be identified with nameplates of nonferrous metal, with nomenclature etched on the nameplate.

3.8 Government furnished property. The gasoline engine (Model 2A016) will be furnished by the Government (see 6.9).

3.9 Conversion instructions. Instructions for converting from engine type compressor drive to electric compressor drive shall be supplied. In addition, a complete list of all replaceable parts (common hardware excepted) showing contractor's part No., part manufacturer's No. and name shall be furnished. Replacement parts for the same or different model end items shall not have the same part No. unless they are interchangeable.

3.10 Workmanship. Sheet metal or structural sections used in fabrication of the refrigeration units shall not be kinked or bent. Bolts, nuts, and screws shall be drawn tight and so secured that they will not work loose under vibration.

3.11 Welding. The surfaces of parts to be welded shall be clean and free of scale or other foreign matter. All scale or flux deposits shall be removed from the finished welds. All external welds at joints or seams in the final assembly shall be finished so that adjacent surfaces shall match.

3.12 Dissimilar metals. Contact areas between dissimilar metals shall be insulated to minimize galvanic corrosion by coating the surfaces of both metals with zinc chromate or asphaltic-type compounds or by strips of rubber conforming to MIL-E-14328. Dissimilar metals are defined in MIL-STD-889.

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3.13 Maintainability. All maintainability requirements for the gasoline engine shall be as specified in MIL-STD-1227. All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement without removing any major component adjustment, repair, or replacement and shall be fastened with easily operated, quick-disconnect fasteners or with screws.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3 and 6.3).
2. Quality conformance inspection (see 4.4).

4.3 First article. When a first article is required, the refrigeration unit shall be tested as specified in table I. The first article shall also be examined for defects listed in table II, and for dimensions and weights specified. Failure to pass any of the tests specified in table I shall be cause for rejection of the first article. The presence of any defects listed in table II or any dimensions or weights not within specified requirements shall be cause for rejection of the first article.

* 4.3.1 Initial production sample inspection. When an initial production sample is required, it shall be examined for the defects in table II, dimensions and weights specified, and tested as specified in test paragraphs 4.6.1, 4.6.2, 4.6.3.2, 4.6.10, and 4.6.11. The presence of any defects or any dimensions or weight not as specified, or failure of any test shall be cause for rejection of the initial production sample.

* 4.3.2 Comparison sample inspection. When a comparison sample is required, it shall be examined for the defects in table II, dimensions and weights specified and tested as specified in test paragraphs 4.6.1, 4.6.2, 4.6.3.2, 4.6.10 and 4.6.11. The presence of any defect or any dimension or weight not as specified, or failure of any test shall be cause for rejection of the comparison sample.

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TABLE I. Testing order list

Test sequence	Requirement paragraph	Test paragraph	Test	Test units
1	3.5	4.6.1	Operation test	All units
2	3.5.2	4.6.2	48 hour test	First article or initial production unit and 1 out of 25 units
3	3.5.2.1	4.6.3.2	0°F (-18°C) refrigeration	First article or initial production unit
4	3.5.2.1.2	4.6.3.3	35°F (2°C) refrigeration	First article or initial production unit
5	3.5.4.14	4.6.4	Defrost test	First article or initial production unit
6	3.5.4.5	4.6.5.1	High temp test	First article or initial production unit
7	3.5	4.6.6	Drop test	First article or initial production unit
8	3.5	4.6.7	Rainfall test	First article or initial production unit
9	3.5.3.1	4.6.8	"D" ring tests	First article or initial production unit
10	3.5.1	4.6.9	Conversion test	First article or initial production unit
11	3.5.4.18	4.6.10	Ref. moisture test	All units
12	3.5.4.1	4.6.11	Ref. leak test	All units
13	3.6	4.6.12	Electromagnetic test	First article or initial production unit
14	3.5	4.6.13	Storage test	First article or initial production unit
15	3.5.2.2	4.6.14	Noise level test	First article or initial production unit

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4.4 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.4.2 In-process inspection. Examination shall be made to establish that no deviation is made from the following requirements. When noncompliance is noted, correction shall be made to the affected items and to the operation.

(a) Installation of insulation (see 3.5.3.2).
(b) Wood preservative treatment (see 3.5.3.2).
(c) Sealing of holes through heat shield (see 3.5.3.2).
(d) Fungus and moisture resistance treatment of electrical components (see 3.5.4.19).
(e) Cleaning, conditioning, priming, and painting (see 3.5.4.20).
(f) Soldering and welding (see 3.5.4.1 and 3.11 respectively).
(g) Insulation of contact areas between dissimilar metals (see 3.12).
(h) Copper-8-quinolinolate is added to preservative as specified (see 3.4.3).

4.4.3 End product inspection. The inspection lot shall be all refrigeration units of one type offered for inspection at one time. The sample unit shall be one completely fabricated refrigeration unit.

4.4.3.1 Visual examination. Examination shall be made of the refrigeration units for the defects listed in table II. The inspection level shall be II with acceptable quality levels (AQL's) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE II. Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Not finished where required	X	
	Color not as specified		X
	Component or assembly painted or coated where not required, for example, paint used in evaporator section	X	

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TABLE II. Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Design, construction and workmanship (applicable to all components and assemblies)	Any component not of specified material	X	
	Any component fractured, split, punctured, dented, deteriorated, bowed, or malformed	X	
	Any component not accessible for servicing where required	X	
Refrigeration unit inclosure	Access doors and panels not located as specified	X	
	Panels not removable or doors not hinged	X	
	No barrel bolts on doors		X
"D" rings	Missing, more or less than specified	X	
Guards	No guards for evaporator and condenser coils	X	
	Not removable		X
Compressor	No provision for removal by use of captive nuts	X	
	No provision for observation of crankcase oil level	X	
Gages	Not provided, not mounted, located or connected as specified		X
Fuel tank	Filler neck, chain secured filler cap, or air vent not provided	X	
	Not located where specified	X	
	Not mounted as specified	X	
	Drain fitting not provided		X
Condenser	Not removable	X	
	Not removable from the front of the unit		X

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TABLE II. Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Lubrication	Fittings not painted red		X
Marking for identification	Missing, incomplete, illegible, not in proper location, not affixed or marked in prescribed manner, not color or number coded		X
Marking (instruction)	Missing, incomplete, illegible, not affixed or marked in prescribed manner	X	

4.4.3.2 Dimensional and weight examination. Examination shall be made of the refrigeration units for defects in dimensions and weight. Any dimension, or weight, not within specified requirements shall be classified a defect. The inspection level shall be S-1 with an AQL of 4.0 expressed in terms of defects per hundred units.

4.4.3.3 End item testing. The end item shall be tested as specified in table I and in the order listed in table I and failure to pass any test shall be cause for rejection. The test specified in 4.6.2 shall be performed on one production unit, selected at random, out of every 25 units or one unit from the lot produced if less than 25, and failure to pass this test shall be cause for rejection of the lot.

4.4.4 Code and standards compliance. Proof of the compliance with the requirements of UL, as applicable to 3.3.1, shall be submitted to the Government representative.

4.5 Packaging inspection. The sampling and inspection of the preservation-packaging, packing and container marking shall be in accordance with MIL-R-3593.

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4.6 Tests.

4.6.1 4-hour operating test. The refrigeration unit evaporator shall be clamped to an inclosure. All refrigeration unit components shall be in place, and the refrigeration unit doors shall be in a normally closed position. The refrigeration unit shall be operated for 4 hours to determine that all mechanical equipment is functioning as intended and all controls are in calibration. The unit shall be stopped and immediately restarted every 30 minutes during operation. The by-pass valve, crankcase pressure regulator shall be checked for proper operation. Prior to the completion of the 4-hour test, the airflow to the condenser shall be blocked off and the unit operated to verify the high pressure cutoff setting. Malfunction of mechanical equipment or improper operation of any controls and safety switch shall constitute failure of this test.

4.6.2 48-hour operating test. A 48-hour operating test shall be conducted at the contractor's plant as follows:

(a) The refrigeration unit shall be clamped to the evaporator opening of the calorimeter for preproduction testing or other insulated structure for other testing.

(b) A combined heat load of 2,000 Btuh (586 W) shall be imposed on the unit.

(c) The refrigeration unit shall be operated 24 hours under the rating conditions of 3.5.2.1.1 and 24 hours under the rating condition of 3.5.2.1.2.

During this test, pressure and temperatures shall be recorded on automatic recording instruments and, upon completion of the test, this data shall be made available to the contracting officer. During this test the type I unit shall be stopped every 2 hours, the amount of oil shall be checked and the unit shall be restarted within 10 minutes. Normal maintenance shall be performed during this test. Inability of the unit to operate or failure to maintain rated conditions for 48 hours due to other than gasoline engine failure shall be cause for failure of this test. Engines which fail will be repaired or replaced and the test continued.

4.6.3 Capacity tests.

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4.6.3.1 Calorimeter calibration. The insulated calorimeter calibration shall be run once prior to capacity tests to establish the coefficient of heat leakage. The coefficient of heat leakage to be used for capacity calculation shall not exceed 25 Btu per sq. ft. per degree F (142 $W/m^2.K$) and the volume of the calorimeter shall not be less than 70 cubic feet (2 m^3). A refrigeration unit shall be bolted to the evaporator opening in the calorimeter and the overall heat leakage coefficient shall be determined as follows:

(a) An electric fan, of capacity to insure that no uncirculated air pocket will exist, shall be installed inside the calorimeter. The electrical energy of the fan motor shall be measured by a watt-meter and its heat equivalent shall be added to other sources of heat added to the calorimeter.

(b) Four thermocouples, shielded from the effects of radiant heat, shall be installed inside the calorimeter and positioned within 6-inches (152 mm) of the ceiling and floor at two opposite corners. The interior temperature shall be taken as the average of the four thermocouples.

(c) Heating elements, which may be incandescent lamp bulbs coupled two in series to cut down on the voltage to minimize radiation, shall be installed approximately in the center of the calorimeter. Electrical heat input shall be measured by a watt hour meter.

(d) Eight thermometers or thermocouples, shielded from the effects of the radiant heat, shall be installed approximately 18-inches (457 mm) from each corner in the room in which the calorimeter is placed. The ambient temperature shall be taken as the average of these eight temperatures.

(e) All openings in the calorimeter shall be sealed and the calorimeter shall remain undisturbed throughout the test.

(f) The ambient temperature shall then be maintained constant within ± 2°F (± 1°C). Sufficient steady, nonvarying heat shall be added to the interior or the calorimeter to maintain a minimum of 60°F (15.5°C) and a maximum of 100°F (38°C) temperature difference between the interior and exterior of the calorimeter.

(g) The test shall be considered concluded when eight consecutive one half hourly readings indicate that:

1. The ambient temperature has been maintained constant within ± 2°F (± 1°C).

2. The temperature inside the calorimeter has been maintained constant within ± 1°F (± 0.5°C).

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(h) The overall heat leakage coefficient U is given by:

U - Calorimeter heat leakage factor

$$U = \frac{H_1}{(T_1 - T_2)}$$

H_1 = Sum of all electrical heat energies added to the structure in Btuh (W)

T_1 = Average interior temperature of insulated structure (100°F (38°C) to 155°F (68°C))

T_2 = Average ambient temperature, ($T_1 - T_2$ shall be in the range of 60°F (15°C) to 100°F (38°C))

(i) Failure to meet the heat leakage coefficient of 25 Btuh per sq. ft. per degree F (142 $W/m^2.K$) shall be cause to reject the test calorimeter.

4.6.3.2 0°F (-18°C) refrigeration test. The capacity of the refrigeration unit for low temperature shall be determined by the method described below to determine compliance with 3.5.2.1.1. Inability of the unit to meet the requirements shall constitute failure of the test. Recorded data shall be required and shall include all specified temperature readings, electric heat input, suction and discharge pressures, and complete calculations. All panels and doors of the refrigeration unit shall be in the normally closed position.

(a) All accessories and appurtenances shall be in their proper place.

(b) Thermocouples, shielded from the effects of radiant heat, shall be installed in the air streams entering the evaporator and leaving the evaporator fan.

(c) Shielded thermocouples shall be located at the approximate center of four equal areas of the condenser face. The condenser inlet temperature will be taken as the average of these four temperatures.

(d) Recording pressure gages shall be used for the compressor suction and discharge pressures.

(e) The procedures described in 4.6.3.1 (b), (c), (d), and (e) shall be followed for this test.

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(f) The refrigeration unit shall then be operated with air entering the condenser at 110°F (43°C), ± 2°F (± 1°C), and sufficient steady, nonvarying heat shall be added to the interior of the calorimeter to maintain 0°F (-18°C) temperature of air entering the evaporator, ± 1°F (± 0.5°C). A watt-hour meter shall be used to indicate the quantity of electrical energy added. Air velocity shall not exceed 100 feet per minute (0.5 m/s) at any point 3 feet (9 m) from the refrigeration unit.

(g) The test shall be considered concluded when eight consecutive 1/2 hourly readings indicate that:

1. The average calorimeter ambient temperature, T_3 has been maintained at steady state of ± 2°F (± 1°C) at a temperature of 100 to 110°F (38° to 43°C).

2. Return air temperature, T_4, entering the evaporator has been maintained at 0°F (-18°C), ± 1°F (± 0.5°C).

3. The net refrigerator capacity shall be considered as the sum of the electrical energy, in Btuh, added during this test plus the heat leakage gain in Btuh, as calculated from the calibration specified in 4.6.3.1. The net refrigeration capacity is given by:

$$Q = U (T_3 - T_4) + H_2$$

U = Overall heat leakage coefficient Btu as determined by paragraph (h) in 4.6.3.1.

H_2 = Sum of all heat energy added to the interior of the calibrated box in Btuh (see (f) above).

4.6.3.3 35°F (1.6°C) refrigeration test. The procedure specified in 4.4.3.1 shall be followed with the following exceptions to determine compliance with 3.5.2.1.2. Inability to meet the requirements shall constitute failure of the test.

(a) In paragraph 4.6.3.1 (f), sufficient heat shall be added to maintain 35°F (1.6°C) temperature, ± 1°F (± 0.5°C), of air entering the evaporator in place of 0°F (-18°C) temperature specified.

(b) In paragraph 4.6.3.1 (g) 2, temperature of the air entering the evaporator has been maintained at 35°F (1.6°C) ± 1°F (± 0.5°C) in place of 0°F (-18°C) temperature specified.

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4.6.4 Defrost test. The refrigeration unit shall be attached to a refrigerator and operated under the low temperature conditions of 3.5.2.1.1 (temperature of air entering the evaporator, 0°F (-18°C)). The evaporator shall be frosted up by spraying water on the coil surface. After the coil is frosted, the refrigeration unit shall be set on the defrost cycle and the condensate collected and weighed to determine compliance with the requirements of 3.5.4.14. Inability to operate through a complete defrost cycle and return to normal refrigeration cycle, to meet condensate weight, overloading the prime mover, splashing or dripping water inside the refrigerator, damaging the compressor because of lack of lubrication or because of refrigerant slugging shall constitute failure of this test.

4.6.5 Ambient temperature tests.

4.6.5.1 High temperature test. With the unit remaining in position after completion of the capacity tests, the unit shall be heat soaked for 6 hours in an ambient of 125°F (52°C), ± 5°F (± 3°C). The unit shall then be started and operated for 4 hours with its control set at 35°F (1.6°C), and for 4 hours with its control set at 0°F (-18°C). Test records of time, pressures, temperatures, and electric load, taken at 1/2 hour intervals shall be required. On the type I units, a thermocouple shall be inserted in the oil sump through the front drain plug opening to a depth of 1-inch in order to measure the crankcase oil temperature. The temperature shall be recorded at three consecutive 30-minute intervals to determine compliance with 3.5.4.5. Any one reading exceeding 250°F (121°C) shall constitute failure of this test. The exhaust gas back pressure shall be measured by attaching a water manometer to the exhaust pipe at a location not more than 2 inches (51 mm) from the engine exhaust manifold flange. Pressures shall be recorded at three consecutive 30-minute intervals to determine compliance with 3.5.4.5.2. Any one reading exceeding 8 inches (203 mm) shall constitute failure of this test. The spark plug head temperature shall be recorded at three consecutive 30 minute intervals to determine compliance with 3.5.4.5. Any one reading exceeding 475°F (246°C) shall be cause for rejection of the unit.

4.6.6 Drop test. While operating, the refrigeration unit shall be raised 1-inch (25 mm) clear from floor and dropped onto a concrete surface fifty times to determine compliance with 3.5. The unit shall be operated for 2 hours. Inability to operate, any damage, refrigerant leaks, distortion or malfunction shall constitute failure of this test.

4.6.7 Rainfall test. Fresh water shall be sprayed on the unit at a rate of 3-inches (76 mm) per hour at an average downgrade angle of 45 degrees to simulate rainfall against the exposed sides of the unit. The unit shall be exposed to the spray for 30 minutes while idle, and then shall be started and operated for 30 minutes while the spray continues. Inability of the unit to start and operate shall constitute failure of the test.

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4.6.8 "D" ring tests. The refrigeration unit shall be lifted free by each "D" ring singularly and then lifted free so that each "D" ring supports one-half the weight to determine compliance with the strength and vertical plane requirements of 3.5.3.1. Inability of a single "D" ring to lift the unit or inability of the unit to be lifted in a vertical plane with both "D" rings shall constitute failure of the test.

4.6.9 Conversion test. The type 1 refrigeration unit shall be converted from gasoline-engine-drive to electric-motor-drive and then operated 2 hours, 1-hour at 0°F (-18°C) and 1-hour at 35°F (1.6°C), to determine compliance with 3.5.1. Inability of the unit to operate at the above temperatures for the time specified shall constitute failure of this test.

4.6.10 Refrigeration moisture test. After each refrigeration unit has been operated in an ambient temperature of 90°F (32°C) for at least 1-hour, the liquid and moisture indicator shall be observed, and shall indicate dry in accordance with the liquid and moisture indicator manufacturer's recommended practice. Inability to meet this requirement shall constitute failure of the test.

4.6.11 Refrigerant leak test. The refrigerant leakage test shall be performed immediately after each required drop test (4.6.6) and as a final test on each unit to determine compliance with 3.5.4.1. The test for leakage shall be made with a halide detector or any other commercial method acceptable to the contracting officer. Any leaks of the refrigeration unit shall constitute failure of the test.

4.6.12 Electromagnetic interference control test. When electromagnetic interference control is specified (see 3.6), the first article or first unit of production, as applicable, shall be tested in accordance with MIL-STD-462, to verify compliance with the pertinent provisions of MIL-STD-461. The Government reserves the right to witness tests performed by the contractor or independent testing agency. The contractor shall furnish the contracting officer written certification that the Interference Control Plan, the EMI/EMC Test Plan, the Electromagnetic Test Report and the requirements meet MIL-STD-461.

4.6.13 Storage test. The completed refrigeration unit shall be subjected to a temperature of plus 155°F (68°C) for 4 hours and then to a temperature of minus 65°F (18°C) for 12 hours to determine conformance with the requirements of 3.5. Any nonconformance with 3.5 shall constitute failure of this test.

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4.6.14 Noise limit test. To conform to the requirements of 3.5.2.2 the refrigeration units shall be tested as specified in MIL-Std-1474. The noise limit catagory shall be "D", tables 2 or 3 as applicable. The operators position shall be 10 feet (3 m) in front of the unit and 5 feet (1.5 m) above ground level. The "service" position shall be 2 feet (0.6 m) in front of the control panel and 4 feet (1 m) above the ground level. Failure to conform to the requirements of 3.5.2.2 shall constitute failure of this test.

5. PACKAGING

5.1 Packaging requirements. The requirements of preservation-packaging, packing and marking with the levels as specified (see 6.2) shall be in accordance with MIL-R-3593.

6. NOTES

6.1 Intended use. Refrigeration units covered by this specification are intended for field use with a portable refrigerator conforming to MIL-R-43024 to maintain a selected storage temperature from 0°F to 35°F (-18°C to 1.6°C).

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of refrigeration unit required (see 1.2).
(c) First article. When a first article sample is not required (see 3.1, 4.3, and 6.4).
(d) When color of finish is other than specified (see 3.4.8).
(e) Source of delivery of Government furnished property (see 3.5.4.5).
(f) When electrical characteristics are other than specified (see 3.5.4.6).
(g) When lubrication instruction fittings and instructions are not required (see 3.5.4.21.2 and 3.5.4.21.5).
(h) Selection of the applicable levels of preservation and packaging and packing (see 5.1 and 5.2).
(i) When engineering design conference is not required (see 6.6).
(j) When drawings are not required (see 6.6).

6.2.1 Contract data requirements. Any requirements for equipment manuals for the item covered by the specification should be included in DD Form 1423, Contract Data Requirements List, and cited in the contract.

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample, initial production item or other specific item described under the definition of a first article in the ASPR. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

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6.4 Supplier's certificate. In a letter notifying the contracting officer that the preproduction sample is ready for inspection and test, the contractor shall be required to certify that satisfactory preliminary operating and performance tests of the preproduction sample have been made and that this item has been manufactured or assembled under the same management and the same or similar manufacturing methods and facilities as will be employed on the production items. Evidence of accepted purchase orders for components, fabrication or assembly by subcontractors and contractor in sufficient detail to demonstrate compliance with this specification, and inspection of preliminary test data, may be requested by the contracting officer.

6.5 Engineering design conference. Within 10 days after award of contract unless otherwise specified (see 6.2), and prior to any work on the preproduction sample including ordering of materials, the contracting officer should arrange an engineering design conference to discuss any questions concerning the details of Government drawings, instructions, and technical requirements of this specification. This conference is to assist the contractor in interpretation of this specification, and agreements made shall neither obligate the Government to accept equipment that is not in accordance with this specification nor relieve the contractor of his obligation to supply the equipment specified.

6.6 Drawings. Unless otherwise specified (see 6.2), the contractor should provide the contracting officer with one complete set of legible prints of all assembly and detail production drawings necessary for the fabrication, assembly, and finish of the refrigeration units covered by this specification. These drawings shall also include a "material list" or, "bill of materials" column on the applicable drawing(s). The contracting officer shall use these drawings as a guide in the inspection and test of the preproduction sample. In the event changes are indicated by the performance of preproduction sample, the production drawings shall be revised to reflect these changes. Within 30 days after completion of a contract, and prior to final payment by the Government, the contractor shall forward to the contracting officer one set of tracings that can be used for reproduction by white print or blueprint method, for the permanent recording of the "as built" design of the refrigerating unit furnished under the contract.

6.7 Conversion kit. The contractors should demonstrate the conversion kit on the preproduction or first unit of production, as applicable (see 3.5.1 and 4.4.9).

6.8 Government furnished property. The component listed below will be furnished to the contractor by the Government:

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Item	Quantity	Nomenclature	Model
1	1	Engine, Gasoline: Air Cooled, 3BHP, 4-Cycle, Military Design	2A016

6.9 Initial production test. When an initial production test sample is required, the contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the initial production test sample.

6.10 Inspection comparison test. When a comparison test sample is required, the contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the comparison test sample.

6.11 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.4).

6.12 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modification, corrections and deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - YD
Air Force - 99

Preparing activity:

Army - GL

Project No. 4110-0206

Review activities:

Army - MD, EL, ME
Navy - MC
DLA - GS

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DOCUMENT IDENTIFIER (Number) AND TITLE

MIL-R-43031E - Refrigeration Units, Mechanical, Panel Type, 3000 BTU/HR (879 W) CAPACITY

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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